Unit 5: Inventory Control (6 Hrs.)

Inventory classification, Different cost associated to Inventory, Economic order quantity, Inventory models with deterministic demands, ABC analysis.

#Past Questions

- **Q8 (a).** Short note: EOQ. -2021
- **Group C Q10.** Define various costs related to inventory. Explain ABC analysis with better example. -2021
- **Q3.** Explain ABC inventory planning system. An auto industry purchases spark plugs at the rate of Rs 25 per piece. The annual consumption of spark plug is 18000 numbers, if the ordering cost is Rs.250 per order and carrying cost is 25% p.a., what would be the EOQ ? If the supplier of spark plugs offers a discount of 5% for order quantity of 3000 numbers per order, do you accept the discount offer? -2024

Introduction to Inventory:

The resources (or items) kept for future use are called as inventory. The resources may be of any type goods, men, machines etc. The problem of Inventory generally arises at the shops (where some items are sold) or factories (where items are manufactured and raw material is required). Although inventory is an idle resource but it is essential to maintain some inventories for smooth functioning of the enterprise. The problem related to how much inventory should be maintained and how much order should be placed is called as Inventory Control Problem.

Inventory control is the process of efficiently overseeing the constant flow of units into and out of an existing inventory. This includes controlling the ordering, storage, and use of components that a company uses in the production of the items it sells.

Need of Inventory

Inventory defend the enterprise against two type of loss, current loss and future loss. It can be easily explained with the help of following example:

A person goes to a shop to purchase an item but that was not in stock then he returns empty hand from the shop it causes the current loss for shopkeeper. If the person returns empty handed twice or thrice from the shop then he could think the shop is not good and he can move to another shop which causes the loss of goodwill of customer as well as the loss of money in future also. To avoid such kind of situations inventory is maintained.

Inventories are carried by organizations because of the following major reasons :

- **Improve customer service-** An inventory policy is designed to respond to individual customer's or organization's request for products and services.

- **Reduce costs-** Inventory holding or carrying costs are the expenses that are incurred for storage of items. However, holding inventory items in the warehouse can indirectly reduce operating costs such as loss of goodwill and/or loss of potential sale due to shortage of items. It may also encourage economies of production by allowing larger, longer and more production runs.

- **Maintenance of operational capability-** Inventories of raw materials and work-in-progress items act as buffer between successive production stages so that downtime in one stage does not affect the entire production process.

- **Irregular supply and demand-** Inventories provide protection against irregular supply and demand; an unexpected change in production and delivery schedule of a product or a service can adversely affect operating costs and customer service level.

- **Quantity discount**- Large size orders help to take advantage of price-quantity discount. However, such an advantage must keep a balance between the storage-cost and costs due to obsolescence, damage, theft, insurance, etc.

- **Avoiding stockouts (shortages)**- Under situations like, labor strikes, natural disasters, variations in demand and delays in supplies, etc., inventories act as buffer against stock out as well as loss of goodwill.

1. Inventory Classification

Inventory classification is the process of organizing and categorizing inventory based on its characteristics, usage, and importance in the production and business process. This helps companies manage stock more efficiently, reduce holding costs, and ensure smooth operations.

Types of Inventory

1. Raw Materials

These are the basic, unprocessed materials used to manufacture finished goods. They are the starting point in the production process.

- **Purpose**: Essential for production to begin.
- **Example**:
  - In a **furniture factory**, wood, nails, and glue are raw materials.
  - In a **textile mill**, cotton or wool is raw material.

2. Work-in-Process (WIP)

WIP refers to items that have entered the production process but are not yet complete. It includes materials, components, and labor costs that are in progress.

- **Purpose**: Shows the value of goods that are midway in production.
- **Example**:
  - In a **car manufacturing plant**, a car with the engine installed but no doors or paint is WIP.
  - In a **bakery**, dough in the oven is WIP before it becomes bread.

3. Finished Goods

These are completed products that are ready for sale or shipment to customers. They are no longer processed and await delivery.

- **Purpose**: Final inventory for fulfilling customer orders.
- **Example**:
  - A **smartphone** packaged and boxed, ready to be sold.
  - A **TV** on display at an electronics store.

4. MRO (Maintenance, Repair, and Operations) Inventory

These are supplies used to support the production process but are not part of the final product. MRO items are essential for keeping machines running and the workplace functional.

- **Purpose**: Support production and maintenance activities.
- **Example**:
  - **Lubricants** for machines.
  - **Cleaning supplies**, safety gloves, and helmets used in factories.

- **Spare parts** used in maintenance.

2. Different Costs Associated with Inventory

1. Ordering Cost

This is the cost incurred **every time an order is placed**, regardless of the order quantity. Frequent ordering can increase these costs, while ordering in bulk may reduce them but increase holding costs.
It Includes:

- Administrative work (e.g., preparing purchase orders)
- Shipping and handling fees
- Communication and processing expenses
- Quality inspection upon arrival

Example:
If a company places 100 orders a year and each order costs $50 to process (paperwork + delivery), the **total ordering cost** is: `100 orders × $50 = $5,000`

Insight:
Frequent small orders increase ordering costs. Businesses try to **balance order frequency and order size** to minimize this cost.

2. Holding / Carrying Cost

This is the cost of **storing and maintaining inventory** over time.
It Includes:

- Warehouse rent or utility bills
- Insurance and security
- Inventory depreciation or obsolescence
- Cost of tied-up capital (money locked in unsold stock)
- Spoilage or damage

Example:
If the value of inventory is $100,000 and carrying cost is 20% annually, then:

```
Carrying cost = $100,000 × 20% = $20,000 per year
```

Insight:
Holding too much inventory increases costs. That's why companies use systems like **Just-in-Time (JIT)** to reduce this.

3. Shortage / Stockout Cost

Cost incurred when inventory is **unavailable** to meet customer demand. It is the penalty cost for running out of stock. It includes the loss of current profit through the sale of item and the loss of goodwill. It also includes the future loss i.e., loss of the profit through sale of item in the future
It Includes:

- Lost sales or backorders
- Customer dissatisfaction or lost loyalty
- Emergency ordering (which may cost more)
- Production stoppage (in manufacturing)

Example:
If a retailer runs out of popular smartphones and loses 50 sales worth $500 each:

```
Lost revenue = 50 × $500 = $25,000
```

Insight:
Too little inventory causes lost business. Businesses must find a balance between avoiding excess and avoiding shortages.

4. Purchase Cost

This is the **actual price paid** to acquire inventory from suppliers.
It Includes:

- Cost per unit of the item
- Bulk discounts (if applicable)
- Taxes and duties (if imported)
- Transportation (if included in item cost)

Example:
If a store buys 1,000 units of a product at $20 each:

```
Total purchase cost = 1,000 × $20 = $20,000
```

Insight:
Companies often negotiate with suppliers for **volume discounts** or better terms to reduce purchase costs.

Summary Table

Cost Type	Explanation
Ordering Cost	Cost per order (paperwork, shipping, processing)
Holding/Carrying Cost	Cost of storing inventory (space, insurance, depreciation)
Shortage/Stockout Cost	Cost of not having enough stock (lost sales, rush orders, dissatisfied customers)
Purchase Cost	Cost to buy the inventory (price per unit, taxes, delivery, etc.)

3. Economic Order Quantity (EOQ)

EOQ is the **optimum order quantity** that minimizes the total inventory cost (ordering + holding costs).

The order quantity specifies the number of units (or quantity) ordered or produced during a single production cycle. The process involved two type of opposite costs, ordering(setup) cost and inventory carrying(holding) cost. The ordering cost decreases with the increase in the size of order whereas inventory carrying cost increases with the increase in the size of order. Therefore, the order quantity must be decided in such a manner that any of these two costs not become too high. The quantity which minimizes the total cost is called as Economic Order Quantity.
The two opposite costs can be shown graphically by plotting them against the order size as shown in Fig. 1.1 below :

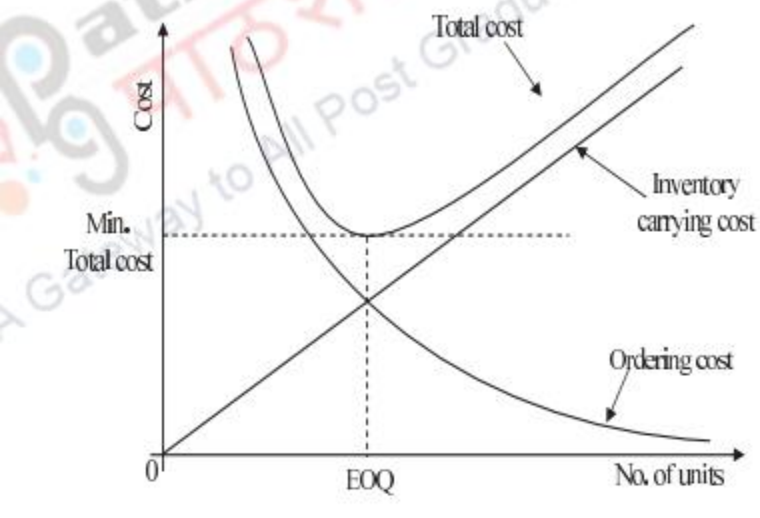


Fig. 1.1: Graph of EOQ

The goal of EOQ is to find a balance between:

- Ordering too frequently (increasing ordering cost), and
- Ordering too much at once (increasing holding cost).

Objective of EOQ:

- Minimize **total cost** = ordering cost + holding cost.

Formula:

$$EOQ = \sqrt{\frac{2DS}{H}}$$

Where:

- D = Annual demand (units)
- S = Ordering cost per order (Rs)
- H = Holding cost per unit per year (Rs)

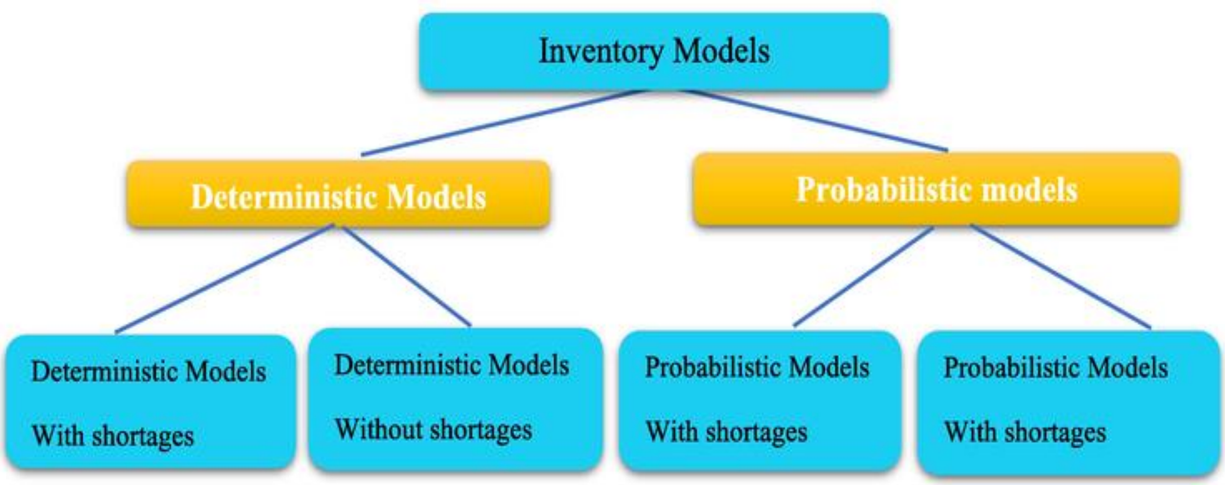
Example:

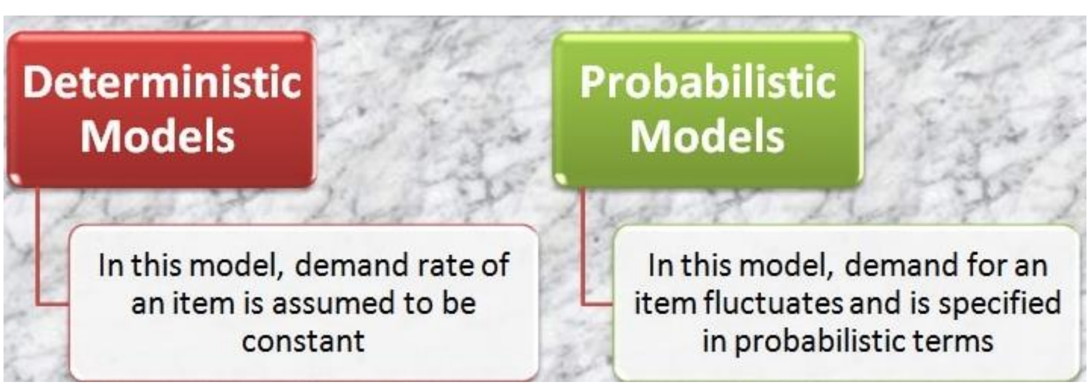
Let's say:

- Annual demand $D = 1000$ units
- Ordering cost $S = Rs.100$
- Holding cost $H = Rs.5$

$$EOQ = \sqrt{\frac{2 \times 1000 \times 100}{5}} = \sqrt{40000} = 200 \text{ units}$$

Interpretation: You should order 200 units each time to achieve the lowest cost.





4. Inventory Models with Deterministic Demand

In **deterministic models**, demand is known and constant over time.

The notation used in the Models

Q = Number of units ordered (supplied) per order

D = Demand in units of inventory per year

N = Number of orders placed per year

TC = Total Inventory cost

C_O = Ordering cost per order

C = Purchase or manufacturing price per unit inventory

C_h = Carrying or holding cost per unit per period of time the inventory is kept

C_s = Shortage cost per unit of inventory

t = The elapsed time between placement of two successive orders

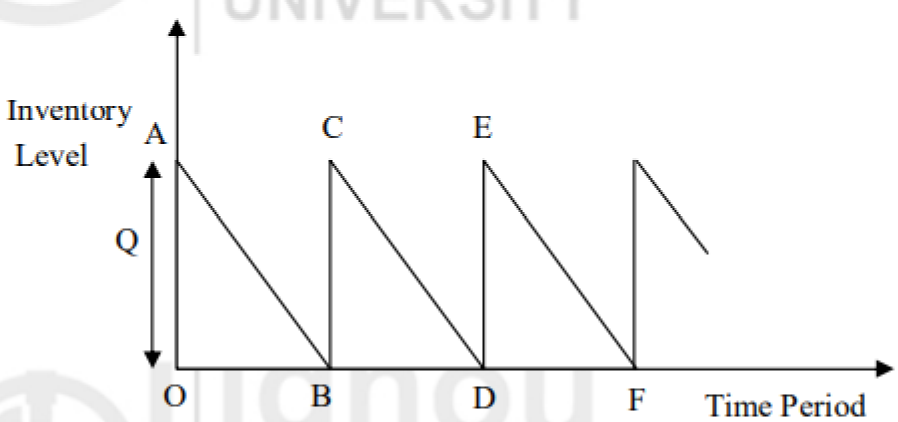
r_p = Replenishment rate at which lot size Q is added to inventory.

EOQ Model with Uniform Demand(Without shortage)

The objective of the EOQ model with uniform demand is to determine an optimum economic order quantity such that the total inventory cost is minimized. We make the following assumptions for this model:

1. Demand rate (D) is constant(uniform) and known;
2. Replenishment rate is instantaneous;
3. Lead time is constant and zero;
4. Purchase price is constant, i.e., discounts are not allowed;
5. Carrying cost and ordering cost are known and constant; and
6. Shortages are not allowed.

The situation can be graphically represented as shown in Fig. below:



The graph in Fig. shows that initially there were Q units in the stock. The number of units goes on decreasing with respect to time to meet the demand and this is represented by the line AB in the graph. When the stock vanishes, i.e., the point B is reached, the stock level rises to Q instantaneously as the replenishment is instantaneous. Since the demand is uniform, the rate of decrease of the quantity remains the same as earlier. Therefore, this is represented by the line CD in the graph, which is parallel to AB. Similarly, EF is parallel to AB and CD due to uniform demand, and so on. Since the demand is uniform, the average inventory is simply the arithmetic mean of the maximum and the minimum levels of inventory. Let Q be the quantity ordered (or replenished) when the minimum level, i.e., zero is reached.

Therefore, the average inventory $= \frac{Q+0}{2} = \frac{Q}{2}$

The ordering cost = Number of orders per year × ordering cost per order

$$= N \times C_O = \frac{D}{Q} \times C_O$$

The carrying cost = Average units in inventory × carrying cost per unit

$$= \frac{Q}{2} \times C_h$$

The total inventory cost is the ordering cost plus the carrying cost. Therefore,

$$TC = \frac{D}{Q} C_O + \frac{Q}{2} C_h$$

The total inventory cost (TC) is minimum at that value of Q where the derivative of TC with respect to Q is zero. Differentiating TC with respect to Q and then equating it to zero, we get

$$-\frac{D}{Q^2} C_O + \frac{1}{2} C_h = 0$$

$$\Rightarrow \frac{D}{Q} C_O = \frac{Q}{2} C_h \quad \text{(i.e., ordering cost = carrying cost)}$$

$$\Rightarrow Q^2 = \frac{2D\,C_O}{C_h} \quad \Rightarrow Q = \sqrt{\frac{2DC_O}{C_h}}$$

This value of Q minimises the total inventory cost (TC) and hence it is the economic order quantity. Let us denote it by Q*. Thus, the EOQ for this model is

$$Q^* = \sqrt{\frac{2DC_O}{C_h}}$$

Hence, the optimum number of orders placed per year $(N^*) = \frac{D}{Q^*}$

The minimum total yearly inventory cost

$$(TC^*) = \frac{D}{Q^*} C_O + \frac{Q^*}{2} C_h = \frac{2DC_O + (Q^*)^2 C_h}{2Q^*}$$

$$= \sqrt{2DC_O C_h} \quad \text{(on simplification)}$$

and the total minimum cost $= (TC^*) +$ cost of material.

Let us consider an example to obtain the total minimum cost in this model.

Example 1: An enterprise requires 1000 units per month. The ordering cost is estimated to be ` 50 per order. In addition to ` 1, the carrying costs are 5% per unit of average inventory per year. The purchase price is ` 20 per unit. Find the economic lot size to be ordered and the total minimum cost.

Solution: We are given that

$$D = \text{Monthly demand} \times 12$$
$$= 1000 \times 12 = 12000 \text{ units per year}$$
$$C_o = ` 50 \text{ per order; } C = ` 20 \text{ per unit}$$
$$C_h = 1 + 5\% \text{ of } ` 20$$
$$= 1 + 1 = ` 2 \text{ per unit of average inventory}$$

The economic lot size, therefore, is given by

$$Q^* = \sqrt{\frac{2DC_O}{C_h}} = \sqrt{\frac{2 \times 12000 \times 50}{2}} = 775 \text{ units}$$

Total minimum cost = TC* + Cost of material

$$= \sqrt{2DC_OC_h} + (12000 \times 20)$$
$$= \sqrt{2 \times 12000 \times 50 \times 2} + 240000$$
$$= \sqrt{2400000} + 240000 = \sqrt{240} \times 100 + 240000$$
$$= 15.5 \times 100 + 240000 = ` 241550$$

Example 2: An auto industry purchases spark plugs at the rate of Rs 25 per piece. The annual consumption of spark plug is 18000 numbers, if the ordering cost is Rs.250 per order and carrying cost is 25% p.a., what would be the EOQ ? If the supplier of spark plugs offers a discount of 5% for order quantity of 3000 numbers per order, do you accept the discount offer

Solution:

Given here,

- Purchase price $C = ₹25$
- Annual demand $D = 18,000$
- Ordering cost $S = ₹250$
- Carrying cost $H = 25\%$ of purchase price = $0.25 \times ₹25 = ₹6.25$
- Discount offer: 5% discount on unit price if order quantity $Q = 3000$

Now

$$EOQ = \sqrt{\frac{2DS}{H}} = \sqrt{\frac{2 \times 18000 \times 250}{6.25}} = \sqrt{\frac{9,000,000}{6.25}} = \sqrt{1,440,000} = 1,200 \text{ units}$$

Again,

🔷 Case A: Without Discount (EOQ = 1200, Unit Price = ₹25)

- Number of orders/year = $\frac{18000}{1200} = 15$
- Ordering cost = $15 \times 250 = ₹3750$
- Average inventory = $\frac{1200}{2} = 600$
- Holding cost = $600 \times 6.25 = ₹3750$
- Purchase cost = $18000 \times 25 = ₹450,000$

$$\text{Total cost} = ₹3750 + ₹3750 + ₹450,000 = ₹457,500$$

🔷 Case B: With Discount (Q = 3000, Unit Price = ₹23.75)

- Discounted price: ₹25 × 95% = ₹23.75
- Number of orders/year = $\frac{18000}{3000} = 6$
- Ordering cost = $6 \times 250 = ₹1500$
- Average inventory = $\frac{3000}{2} = 1500$
- Holding cost = $1500 \times (0.25 \times 23.75) = 1500 \times 5.9375 = ₹8906.25$
- Purchase cost = $18000 \times 23.75 = ₹427,500$

$$\text{Total cost} = ₹1500 + ₹8906.25 + ₹427,500 = ₹437,906.25$$

- **Savings with discount** = 457,500 – 437,906.25 = **₹ 19,593.75**

Conclusion: Ordering **3,000** at the 5% discounted price lowers total annual cost by ~₹ 19.6 thousand. **Accept the discount offer.**

5. ABC Analysis / ABC Inventory System

The **ABC inventory planning system** is a method used to categorize inventory items based on their **value and importance** to a business. It follows the **Pareto Principle (80/20 Rule)**, which states that roughly **80% of a company's sales or costs** come from **20% of its inventory items**. It helps businesses prioritize their inventory items based on **value and importance**, ensuring better control and management of stock. Inventory is divided into **three categories**:

Category	Description	% of Items	% of Value	Management Focus
A	High-value, low-quantity items	**10-20%**	**70-80%**	Tight control, frequent reviews
B	Moderate-value, moderate-quantity items	**20-30%**	**15-20%**	Regular monitoring
C	Low-value, high-quantity items	**50-70%**	**5-10%**	Minimal control, bulk ordering

Purpose of ABC Analysis

- To focus control efforts on the most valuable items.
- To optimize inventory management and reduce costs.
- To improve forecasting, purchasing, and stock decisions.

How to Perform ABC Analysis

1. **List all inventory items** with their annual usage (in units) and unit cost.
2. **Calculate Annual Consumption Value/Total Cost** : (Annual usage × Cost per unit)
3. **Rank items** in descending order of consumption value.
4. **Calculate cumulative totals** and percentages.
5. **Classify into A, B, and C categories** based on cumulative percentages.

ABC analysis example
Here is a working ABC analysis example, showing how to divide your inventory using annual consumption value (or usage value).

Q. Perform the ABC Analysis using the following data.

Item	Units	Unit Price (Rs.)
1	700	50
2	2400	30
3	150	100
4	60	220
5	3800	15
6	4000	5
7	6000	2
8	300	35
9	30	80
10	2900	4

Calculate Annual Consumption Value/Total Usage : (Annual usage × Cost per unit)

Item	Units	Unit Price (Rs.)	Usage (Rs)	Percentage	Rank
1	700	50	35000	14.04	3
2	2400	30	72000	28.88	1
3	150	100	15000	6.02	5
4	60	220	13200	5.29	6
5	3800	15	57000	22.86	2
6	4000	5	20000	8.02	4
7	6000	2	12600	5.05	7
8	300	35	10500	4.21	9
9	30	80	2400	0.96	10
10	2900	4	11600	4.65	8
		Total Amount	249300		

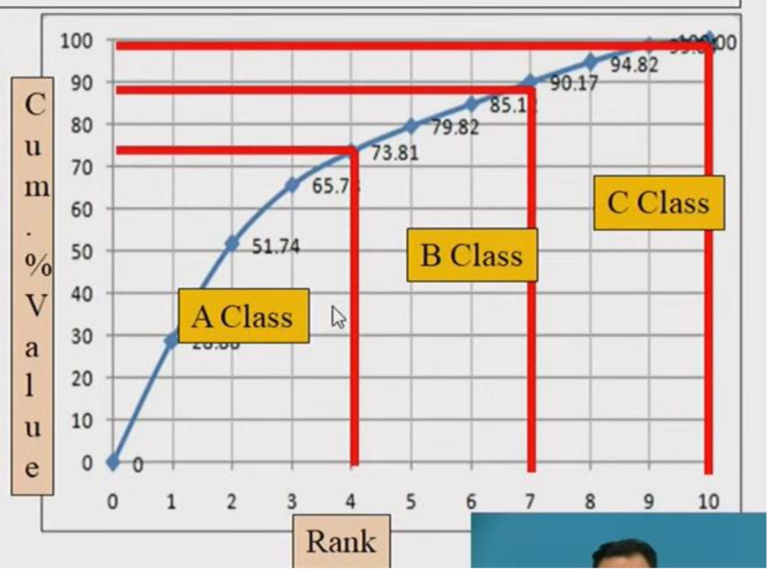
List items in descending order based on Total usage or its percentage(highest to lowest) & calculate cumulative % Value

Item	Rank	Percentage	Cumulative Percentage Value
2	1	28.88	28.88
5	2	22.86	51.74
1	3	14.04	65.78
6	4	8.02	73.81
3	5	6.02	79.82
4	6	5.29	85.12
7	7	5.05	90.17
10	8	4.65	94.82
8	9	4.21	99.04
9	10	0.96	100.00

Classify into A, B, and C categories based on cumulative percentages and represent in graph.

Graphical Presentation

	Item	Rank	Cumulative Percentage Value
A Class	2	1	28.88
	5	2	51.74
	1	3	65.78
	6	4	73.81
B Class	3	5	79.82
	4	6	85.12
	7	7	90.17
C Class	10	8	94.82
	8	9	99.04
	9	10	100.00



Importance of ABC Analysis

ABC analysis helps businesses **prioritize inventory management** by categorizing items into **A (high-value), B (moderate-value), and C (low-value)** groups. This ensures efficient resource allocation and cost control.

- **Increased Inventory Optimization:** The analysis identifies the products that are in demand. A company can then use its precious warehouse space to adequately stock those goods and maintain lower stock levels for Class B or C items.
- **Improved Inventory Forecasting:** Monitoring and collecting data about products that have high customer demand can increase the accuracy of sales forecasting. Managers can use this information to set inventory levels and prices to increase overall revenue for the company.

- **Better Pricing:** A surge in sales for a specific item implies demand is increasing and a price increase may be reasonable, which improves profitability.
- **Informed Supplier Negotiations:** Since companies earn 70% to 80% of their revenue on Class A items, it makes sense to negotiate better terms with suppliers for those items. If the supplier will not agree to lower costs, try negotiating post-purchase services, down payment reductions, free shipping or other cost savings.
- **Strategic Resource Allocation:** ABC analysis is a way to continuously evaluate resource allocation to ensure that Class A items align with customer demand. When demand lowers, reclassify the item to make better use of personnel, time and space for the new Class A products.
- **Better Customer Service:** Service levels depend on many factors, like quantity sold, item cost and profit margins. Once you determine the most profitable items, offer higher service levels for those items.
- **Better Product Life Cycle Management:** Insights into where a product is in its life cycle (launch, growth, maturity or decline) are critical for forecasting demand and stocking inventory levels appropriately.
- **Control Over High-Cost Items:** Class A inventory is closely tied to a company's success. Prioritize monitoring demand and maintaining healthy stock levels, so there's always enough of the key products on hand.
- **Sensible Stock Turnover Rate:** Maintain the stock turnover rate at appropriate levels through methodical inventory control and data capture.
- **Reduced Storage Expenses:** By carrying the correct proportion of stock based on A, B or C classes, you can reduce the inventory carrying costs that come with holding excess inventory.
- **Simplified Supply Chain Management:** Use an ABC analysis of inventory data to determine if it's time to consolidate suppliers or shift to a single source to reduce carrying costs and simplify operations.

Disadvantages of ABC Analysis

1. **Too Simplistic**

- Only considers **one variable** (e.g., sales value), ignoring demand variability, sales frequency, etc.
- Lacks granularity—hundreds of items in **C category** may need different handling.

2. **Static & Outdated**

- Market trends can quickly shift items between categories (e.g., a **C item** suddenly becoming high-demand).
- Requires **frequent updates** to remain accurate.

3. **Time-Consuming**

- Regular reclassification is needed, which can divert focus from **actionable inventory decisions**.

Key Takeaway

ABC analysis is a **powerful tool for inventory prioritization** but must be **used dynamically** and combined with other metrics (like demand variability) for optimal results. Regular reviews prevent misclassification and stock issues.